Austin Hoover

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Education and Employment

Accelerator Physicist, Oak Ridge National Laboratory 2024–

Postdoctoral Research Associate, Oak Ridge National Laboratory 2022–2024

Ph.D. in Physics, University of Tennessee 2018–2022

B.S. in Physics, Wheaton College (IL) 2014–2018

Journal Articles

Demonstration of a novel phase space painting method in a coupled lattice to mitigate space charge in high-intensity hadron beams. N. Evans, A. Hoover, T. Gorlov, and V. Morozov. Phys. Rev. Letters 136, 075002. [link] [arxiv] [data] 2026

N-dimensional maximum-entropy tomography via particle sampling. A. Hoover. Phys. Rev. Accel. Beams 28, L084601 (Letter, Editor's Choice). [link] [arxiv] [code] 2025

Four-dimensional phase space tomography from one-dimensional measurements of a hadron beam. A. Hoover. Phys. Rev. Accel. Beams 27, 122802. [link] [arxiv] [data] 2024

High-dimensional maximum-entropy phase space tomography using normalizing flows. A. Hoover and J.C. Wong. Phys. Rev. Research 6.3, 033163. [link] [arxiv] [code] 2024

Analysis of a hadron beam in five-dimensional phase space. A. Hoover, K. Ruisard, A. Aleksandrov, A. Zhukov, and S. Cousineau. Phys. Rev. Accel. Beams 26.6, 064202. [link] [arxiv] [data] 2023

Four-dimensional emittance measurement at the Spallation Neutron Source. A. Hoover and N. Evans. Nucl. Instrum. Methods Phys. Res., Sect. A 1041, 167376. [link] [arxiv] 2022

Computation of the matched envelope of the Danilov distribution. A. Hoover, N. Evans, and J. Holmes. Phys. Rev. Accel. Beams 24.4, 044201. [link] 2022

Conference/Workshop Proceedings

Invited Talks

Eigenpainting for mitigation of space charge in high-intensity hadron beams. International Particle Accelerator Conference. 2027

High-dimensional maximum-entropy phase space tomography. North American Particle Accelerator Conference. [paper] [slides] 2025

Contributed Talks

Designing experiments to study large space charge tune shifts in the SNS accumulator ring. ICFA Mini-Workshop on Space Charge. [slides] 2026

Four-dimensional phase space tomography from one-dimensional measurements of a hadron beam. International Linear Accelerator Conference. [slides] [code] 2026

Bunch compression experiments at SNS. International Muon Collider Collaboration Annual Meeting. [slides] 2026

Repurposing existing accelerator facilities for Muon Collider R&D. APS Global Physics Summit. [slides] 2026

Proton Driver for the Demonstrator and Muon Collider. US Muon Collider Annual Meeting. [slides] 2025

Bunch rotation experiments and benchmarking at SNS. International Muon Collider Collaboration Meeting. [slides] 2025

SNS accelerator synergies for muon collider R&D. Princeton Muon Collider Organizational Workshop. [slides] 2024

The impact of high-dimensional phase space correlations on the beam dynamics in a linear accelerator. ICFA Workshop on High-Intensity and High-Brightness Hadron Beams. [paper] [slides] 2023

Measurements of the five-dimensional phase space distribution of an intense ion beam. North American Particle Accelerator Conference. [paper] [slides] 2022

Accelerator research and development at the Spallation Neutron Source. Particle Accelerators: Technology and Applications (UTK graduate course). [slides] 2022

Self-consistent, angular-momentum-dominated hadron beams for space charge mitigation. ICFA Mini-Workshop on Space Charge. [slides] 2022

Development of an injection-painted self-consistent beam at the Spallation Neutron Source. ICFA Workshop on High-Intensity and High-Brightness Hadron Beams. [paper] 2021

Computation of the matched envelope of the Danilov distribution. APS April Meeting. [slides] 2021

Posters

Search for Vlasov-Poisson equilibrium distributions using generative models. A. Hoover. International Particle Accelerator Conference. [paper] 2026

Feasibility of proton bunch compression in an operational high-power accelerator. A. Hoover, V. Morozov, C. Piller, A. Narayan. North American Particle Accelerator Conference. [paper] 2025

MENT-Flow: maximum-entropy tomography using normalizing flows. A. Hoover, C. Wong. International Particle Accelerator Conference. [paper] 2024

Detailed characterization of a five-dimensional phase space distribution. A. Hoover, K. Ruisard, A. Aleksandrov, S. Cousineau, A. Zhukov. International Particle Accelerator Conference. [paper] 2023

Simulation of 4D emittance measurement at the Spallation Neutron Source. A. Hoover, N. Evans. International Particle Accelerator Conference. [paper] 2021

Coauthor

Beam loss modeling and mitigation due to intra-beam stripping. S. Kakkar, V. Morozov, L. Lee, N. Evans, A. Zhukov, A. Hoover. North American Particle Accelerator Conference. [paper] 2025

Exploring space charge mitigation with eigenpainting at the SNS. N.J. Evans, A. Hoover, T. Gorlov, V. Morozov. North American Particle Accelerator Conference. [paper] 2025

Progress toward halo modeling at the SNS Beam Test Facility. K. Ruisard, A. Aleksandrov, A. Hoover, T. Thompson, A. Zhukov. Linear Accelerator Conference. [paper] 2024

Proposed muon collider R&D at the SNS. V. Morozov, A. Hoover, D. Stratakis, F. Pilat, N. Evans, S. Cousineau. International Particle Accelerator Conference. [paper] 2024

Measurement and modeling of beam transport in the FODO line of the Spallation Neutron Source Beam Test Facility. T. Thompson, K. Ruisard, A. Hoover, A. Aleksandrov, A. Zhukov. International Particle Accelerator Conference. [paper] 2024

Status of Spallation Neutron Source Beam Test Facility and progress of beam dynamics studies. K. Ruisard, A. Aleksandrov, S. Cousineau, R. Dickson, B. Han, A. Hoover, A. Shishlo, T. Thompson, V. Tzoganis, R. Welton, A. Zhukov. International Particle Accelerator Conference. [paper] 2024

Self-consistent injection painting for space charge mitigation. N. Evans, A. Hoover, T. Gorlov, V. Morozov. ICFA Workshop on High-Intensity and High-Brightness Hadron Beams. [paper] 2023

Measurements at peak operational beam current in the SNS beam test facility. K. Ruisard, A. Hoover, A. Aleksandrov, A. Zhukov, S. Cousineau. International Particle Accelerator Conference. [paper] 2023

Phase space painting of a self-consistent Danilov distribution in the SNS Ring. N. Evans, A. Hoover, T. Gorlov, V. Morozov. International Particle Accelerator Conference. [paper] 2023

Model/measurement comparison of the transverse phase space distribution of an RFQ-generated bunch at the SNS BTF. K. Ruisard, A. Aleksandrov, S. Cousineau, A. Hoover, A. Zhukov. North American Particle Accelerator Conference. [paper] 2022

Observation of current-driven features of 2.5 MeV ion bunch with complete and efficient 5D measurements at the SNS Beam Test Facility. K. Ruisard, A. Aleksandrov, S. Cousineau, A. Hoover, A. Zhukov. International Linear Accelerator Conference. [paper] 2022

Open XAL Status Report 2022. A. Zhukov, A. Hoover, A. Shishlo, J. F. Esteban Müller, E. Laface, Y. Levinsen, N. Milas. International Particle Accelerator Conference. [paper] 2021

Miscellaneous

Six Dimensional Distributions at the SNS Beam Test Facility. K. Ruisard, A. Aleksandrov, A. Hoover. APS Division of Physics of Beams Annual Newsletter. [link] 2022

Towards the production of a self-consistent phase space distribution. A. Hoover. PhD Diss., University of Tennessee. [link] 2022

Seminars

Eigenmode painting in high-intensity accelerators. Accelerator Science and Engineering Traineeship (ASET) Seminar Series, Michigan State University. [link] 2026

Accelerator and beam physics research at ORNL. Adlai E. Stevenson High School Physics Club. 2026

Accelerator research and development at the Spallation Neutron Source. University of Tennessee Nuclear Engineering Department, graduate course (Particle Accelerators: Technology and Applications). 2022

Teaching Experience

Instructor, United States Particle Accelerator School, *Beam Physics with Intense Space Charge.* [website] 2026

Teaching Assistant, United States Particle Accelerator School, *Accelerator Fundamentals.* 2021

Graduate Teaching Assistant, Department of Physics, University of Tennessee. 2018–2019

Teaching Assistant, Department of Physics, Wheaton College. 2018–2019

Service

Program committee, United States Muon Collider Collaboration (USMCC) Annual Meeting 2026

Organizing committee, International Muon Collider Collaboration (IMCC) Annual Meeting 2026

United States Muon Collider Collaboration (USMCC) Leadership Strategy Group (Proton Driver) 2025–

Reviewer for Physical Review Letters (1), Physical Review Accelerators and Beams (2), and Nature Communications Physics (1).